

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |

Kent, OH - US-based, F-1 OPT (open to sponsorship)

PROFESSIONAL SUMMARY

Forward Deployed AI Engineer with shipped production Claude integration and end-to-end agentic systems in regulated healthcare. Built and published Manga Lens to the Chrome Web Store integrating **Claude Sonnet** and 3 other vision providers behind a single interface - production Claude integration discipline, not demo code. Shipped a 5-stage multi-agent healthcare claims pipeline with **schema-validated JSON contracts between agents** (MCP-adjacent pattern) to prevent cascading hallucinations. Healthcare RAG with ~35% retrieval precision gain and >90% grounding alignment via dedicated eval harness. Packaged AI inference as FastAPI/Flask services in Docker (~30% post-deploy defect reduction). Master's in CS (Kent State). US-based, F-1 OPT, open to sponsorship.

CORE COMPETENCIES

Claude API Integration • MCP-Adjacent Multi-Agent Design • Agent Skills & Sub-Agents • Prompt Engineering • Evaluation Frameworks • Deployment at Scale • RAG & Grounding • Python • TypeScript • FastAPI • Docker • Customer-Embedded Delivery

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present | USA

Jr. AI/ML Engineer Trainee - Production Claude / LLM Integration, Healthcare

- Built **agentic LLM workflows** for multi-step healthcare queries (eligibility, care workflow navigation, documentation) with structured reasoning, tool discipline, and grounding rules; improved agent response stability ~**25%**.
- Shipped **RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved contextual retrieval precision ~**35%** and reduced irrelevant retrieval >**30%**. **Eval harness** raised grounded response alignment above **90%**.
- Packaged production LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects ~**30%**.
- Maintained HIPAA-conscious data governance: de-identification pipelines, data lineage, stakeholder-facing system-limitation docs - the documentation discipline white-glove enterprise deployments depend on.
- Shipped predictive ML pipelines (scikit-learn, XGBoost); improved recall **15-20%** on high-risk categories while holding precision >**90%**.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023 | India

Database & DevOps Performance Engineer (Intern) - Compliance-Sensitive Enterprise ERP

- Built **CI/CD pipelines (Jenkins)** for version-controlled SQL and stored-procedure deploys with structured release validation and rollback checkpoints; reduced deployment errors >**30%**, improved release cycle time ~**35-40%**.
- Built **SQL Server DMV + Grafana** performance dashboards to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~**25%**, improved RCA speed ~**30%**.
- Tuned T-SQL stored procedures across Order-to-Cash modules; cut query execution time ~**20%** and data retrieval latency ~**25%**.
- Supported **RBAC + audit logging** for financial modules in a compliance-sensitive oil & gas environment.

Suvidha Foundation

Jan 2022 - Mar 2022 | India

Machine Learning Engineer - Applied NLP, Document Q&A with RAG

- Built a transformer-based summarization system over **5,000+ sessions**; hierarchical long-context summarization cut manual review time **60-70%**.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~**30%**, raised answer accuracy >**85%**.

PROJECTS

Manga Lens - Chrome Extension with Production Claude Sonnet Integration (shipped, Chrome Web Store)

Full-stack browser extension built independently - Manifest V3, content scripts, service workers, captureVisibleTab for panel capture. Integrates **Claude Sonnet** alongside GPT-4o mini, GPT-4.1 Nano, and Ollama/minicpm-v local. Multi-section capture pipeline handles tall webtoon panels via viewport-slice screenshots with coordinate remapping and dedup. Seven-day translation cache, per-domain selector configs for 29 manga/webtoon sites, privacy policy, narrowed host permissions.

Production Claude API integration in user-facing software.

Claude Sonnet, multi-provider AI integration, TypeScript, Manifest V3, service workers

Agentic Healthcare Claims Pipeline - Schema-Contracted Multi-Agent (MCP-Adjacent)

Multi-agent pipeline (Intake -> Validation -> Consistency -> Duplicate Detection -> Fraud Risk) with **schema-validated JSON contracts between agents** - the same tool-discipline pattern MCP server design depends on. RAG-grounded CPT/ICD validation against vector-indexed policy docs, ANN duplicate detection, explainable risk scoring with **audit-ready reasoning traces**.

Multi-agent orchestration, JSON schema contracts (MCP-adjacent), RAG, vector search, audit traces

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack **FastAPI + React/TypeScript/Vite/Tailwind** app coordinating multimodal APIs (Perplexity Sonar + Replicate).

Intermediate structured prompt transformation layers improved contextual alignment **~30%** and first-pass image success **~25-30%**.

FastAPI, TypeScript, structured prompt orchestration

Driver Drowsiness Detection - Real-Time YOLOv8

Replaced a two-stage CNN pipeline with a **unified YOLOv8** model; reduced inference latency **~30%**. Sliding-window aggregation suppressed blink-driven false positives (**~25% reduction**).

YOLOv8, OpenCV, real-time CV

EDUCATION

Master's Degree, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: Python, TypeScript, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite), React, Node.js

LLM & GenAI: Claude API, MCP-adjacent multi-agent design, agent skills, sub-agents, prompt engineering, context engineering, evaluation frameworks, RAG, grounding, embeddings, vector search

Infra & DevOps: Docker, Jenkins, CI/CD, RESTful APIs, observability/logging, Grafana

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex

Enterprise / Compliance: HIPAA-conscious governance, RBAC, audit logging, release validation, rollback checkpoints